

Frequency-Guided Posterior Sampling for Diffusion-Based Image Restoration

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Abstract

Image restoration aims to recover high-quality images from degraded observations. When the degradation process is known, the recovery problem can be formulated as an inverse problem, and in a Bayesian context, the goal is to sample a clean reconstruction given the degraded observation. Recently, modern pretrained diffusion models have been used for image restoration by modifying their sampling procedure to account for the degradation process. However, these methods often rely on certain approximations that can lead to significant errors and compromised sample quality. In this paper, we propose a simple modification to existing diffusion-based restoration methods that exploits the frequency structure of the reverse diffusion process. Specifically, our approach, denoted as Frequency Guided Posterior Sampling (FGPS), introduces a time-varying low-pass filter in the frequency domain of the measurements, progressively incorporating higher frequencies during the restoration process. We provide the first rigorous analysis of the approximation error of FGPS for linear inverse problems under distributional assumptions on the space of natural images, demonstrating cases where previous works can fail dramatically. On real-world data, we develop an adaptive curriculum for our method’s frequency schedule based on the underlying data distribution. FGPS significantly improves performance on challenging image restoration tasks including motion deblurring and image dehazing. The source code is available at <https://github.com/darshanthaker/FGPS>.

1. Introduction

Image restoration is a fundamental problem in computer vision whose goal is to recover high-quality images from degraded observations. Many image restoration problems can be cast as inverse problems, where the task is to recover an underlying signal $\mathbf{x}_0 \in \mathbb{R}^n$ from noisy measurements $\mathbf{y} \in \mathbb{R}^m$ that are obtained from $\mathbf{x}_0$ via

$$\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \mathbf{z}. \quad (1)$$

In the above, $\mathcal{A} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ denotes the forward operator, which we assume to be known, and $\mathbf{z} \in \mathbb{R}^m$ is additive Gaussian noise. In most applications, recovering $\mathbf{x}_0$ is difficult because the forward operator is a many-to-one mapping. To address this challenge, researchers incorporate prior knowledge about $\mathbf{x}_0$, including sparsity [5, 16], low-rank structure [4, 6], total variation [32], or deep generative priors [3, 29]. One way of enforcing this knowledge is by solving a regularized optimization problem where the objective is to find an $\mathbf{x}_0$ that maximizes the likelihood under the chosen prior model while satisfying the constraint $\mathbf{y} = \mathcal{A}(\mathbf{x}_0)$. Another approach is to follow the Bayesian viewpoint where instead of recovering an exact $\mathbf{x}_0$, we sample from $p(\mathbf{x} | \mathbf{y})$ where Equation (1) is the model for $p(\mathbf{y} | \mathbf{x}_0)$ [38]. The advantage of this interpretation is that we can draw multiple plausible samples. In image restoration tasks, the diversity of samples can help the user pick a visually appealing reconstruction.

Recently, diffusion models [20, 36] have shown tremendous success in modeling complex probability distributions, allowing one to generate diverse and visually coherent samples in image, video, and audio tasks [14, 21, 25]. During training, diffusion models learn to reverse a gradual noising process that transforms data $\mathbf{x}_0$ to pure noise $\mathbf{x}_T$ over T steps, where $\mathbf{x}_t$ denotes an intermediate step of the reverse diffusion process. At test time, they generate samples by iteratively denoising random noise, progressively refining it into coherent data. Naturally, these models can also be used to develop samplers for $p(\mathbf{x} | \mathbf{y})$ in inverse problem settings. However, while one can train a problem-specific diffusion model to sample directly from $p(\mathbf{x} | \mathbf{y})$ [33], this requires training new diffusion models for every inverse problem, limiting the practical use of such approaches.

A popular approach which does not require training a new diffusion model for $p(\mathbf{x} | \mathbf{y})$ is to modify the sampling process of a pretrained unconditional diffusion model so that it produces samples that are consistent with the given measurement $\mathbf{y}$ [9, 24, 35, 37, 41]. These methods are based on decomposing the conditional score function $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y})$ as $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$ using the Bayes rule. While the first term is learned by a

基于扩散的图像复原的频率引导后验采样

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摘要

图像修复旨在从退化观测数据中恢复高质量图像。当退化过程已知时，该修复问题可转化为逆问题，在贝叶斯框架下，目标是在给定退化观测数据的情况下采样出清晰的重建结果。近年来，现代预训练扩散模型通过修改采样过程以适应退化过程，被广泛应用于图像修复领域。然而，这些方法常依赖特定近似处理，可能导致显著误差并降低样本质量。本文提出一种基于逆扩散过程频率结构的改进方案，对现有扩散修复方法进行优化。具体而言，我们提出的频率引导后验采样（FGPS）方法，在测量数据的频域中引入时变低通滤波器，通过逐步融合高频成分来实现图像修复。我们首次对自然图像空间的分布假设下线性逆问题的FGPS近似误差进行了严格分析，揭示了先前研究可能失效的典型示例。基于真实数据，我们开发了根据底层数据分布自适应调整方法频率表的课程体系。FGPS在运动去模糊和图像去雾等具有挑战性的图像修复任务中显著提升了性能。源代码可在<https://github.com/darshanthaker/FGPS>获取。

1. 介绍

图像复原是计算机视觉中的一个基本问题，其目标是从退化观测中恢复高质量图像。许多图像复原问题可表述为逆问题，其任务是从噪声测量 $\mathbf{y} \in \mathbb{R}^m$ 中恢复底层信号 $\mathbf{x}_0 \in \mathbb{R}^n$ ，这些测量值是通过 $\mathbf{x}_0$ 获得的。

$$\mathbf{y} = A(\mathbf{x}_0) + \mathbf{z}. \quad (1)$$

在上述公式中， $A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ 表示前向算子（我们假设其已知），而 $\mathbf{z} \in \mathbb{R}^m$ 是加性高斯噪声。在大多数应用中，由于前向算子是多对一映射，恢复 $\mathbf{x}_0$ 非常困难。为解决这一挑战，研究者引入了关于 $\mathbf{x}_0$ 的先验知识，包括稀疏性[5, 16]、低秩结构[4, 6]、全变差[32]或深度生成先验[3, 29]。一种实现该知识的方法是求解正则化优化问题，其目标是在满足约束条件 $\mathbf{y} = A(\mathbf{x}_0)$ 的前提下，找到使先验模型下似然函数最大化的 $\mathbf{x}_0$ 。另一种方法是遵循贝叶斯观点，即我们不是恢复一个精确的 $\mathbf{x}_0$ ，而是从 $p(\mathbf{x}|\mathbf{y})$ 中采样，其中公式(1)是 $p(\mathbf{y}|\mathbf{x}_0)$ 的模型[38]。这种解释的优势在于我们可以抽取多个合理的样本。在图像修复任务中，样本的多样性有助于用户选择视觉上更吸引人的重建结果。

最近，扩散模型[20, 36]在建模复杂概率分布方面取得了巨大成功，使得人们能够在图像、视频和音频任务中生成多样且视觉连贯的样本[14, 21, 25]。在训练过程中，扩散模型学习逆转一个渐进的噪声化过程，该过程将数据 $\mathbf{x}_0$ 转化为纯噪声 $\mathbf{x}_T$ ，经过 T 个步骤，其中 $\mathbf{x}_t$ 表示反向扩散过程的中间步骤。在测试时，它们通过迭代去噪随机噪声生成样本，逐步将其精炼为连贯数据。自然地，这些模型也可用于逆问题设置中开发 $p(\mathbf{x}|\mathbf{y})$ 的采样器。然而，虽然可以训练一个针对特定问题的扩散模型来直接从 $p(\mathbf{x}|\mathbf{y})$ 中采样[33]，但这需要为每个逆问题训练新的扩散模型，从而限制了此类方法的实际应用。

一种无需为 $p(\mathbf{x}|\mathbf{y})$ 训练新扩散模型的流行方法，是修改预训练无条件扩散模型的采样过程，使其生成与给定测量值 $\mathbf{y}$ 一致的样本[9, 24, 35, 37, 41]。这些方法基于将条件评分函数 $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|\mathbf{y})$ 分解为 $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$ ，利用贝叶斯规则。虽然第一项通过

pretrained unconditional diffusion model, the second term is analytically intractable in general because $\mathbf{x}_t$ is the result of a complex denoising process. Thus, in practice, we use approximations for this term, typically by first computing $\mu_{0|t} = \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]$ using the learned diffusion model, known as the Tweedie estimate [17], and following the gradient of a loss that measures the deviation between $\mathcal{A}(\mu_{0|t})$ and $\mathbf{y}$. While intuitive, the approximation error from the Tweedie estimate is difficult to control. Previous work has attempted to address this by designing heuristics that ensure that the modifications to $\mathbf{x}_t$ respect the intermediate data manifold learned by the diffusion model. However, these approaches make strong assumptions about the data manifold's structure, such as linearity, or require that $\mu_{0|t}$ remains on the clean data manifold throughout all iterations [10, 42].

Our work aims to better understand and control this approximation error by analyzing images in the frequency domain. Our main contributions are:

1. We develop a new unsupervised diffusion-based image restoration algorithm, denoted as **Frequency Guided Posterior Sampling (FGPS)**. Our algorithm leverages the observation that diffusion models generate images hierarchically in the frequency domain [30]. Thus, to control the approximation error from estimating $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$, we apply a time-varying low-pass filter to $\mathbf{y}$ which progressively includes higher frequency ranges during the reverse diffusion process.
2. We then perform a precise theoretical analysis of the approximation errors in posterior sampling methods for linear inverse problems under distributional assumptions on the data. Next, exploiting the frequency distribution of natural images, we give precise examples of forward operators where FGPS has small approximation error compared to prior work.
3. In practice, we develop a data-dependent curriculum for the low-pass filters which allows the reverse process to adapt to spectral properties of the data distribution. While being easy to implement, our algorithm leads to a significant boost in performance on challenging linear and nonlinear image restoration problems, such as motion deblurring and image dehazing.

2. Background

2.1. Diffusion Models

Diffusion models are a class of score-based generative models that sample from a given data distribution by learning to reverse a gradual noising process [20, 36]. This noising process, known as the forward process, is represented in continuous time as an Itô Stochastic Differential Equation

(SDE) for $t \in [0, 1]$

$$d\mathbf{x}_t = -\frac{1}{2}\beta(t)\mathbf{x}_t dt + \sqrt{\beta(t)}d\mathbf{w}_t. \quad (2)$$

The SDE is characterized by its drift coefficient $-\frac{1}{2}\beta(t)$ and diffusion coefficient $\sqrt{\beta(t)}$, with $\mathbf{w}_t$ representing the standard Wiener process over the interval $[0, 1]$. The initial condition of this SDE is a datapoint $\mathbf{x}_0 \sim p(\mathbf{x})$. The drift and diffusion coefficients are chosen such that $\mathbf{x}_t$ goes from the clean data distribution at $t = 0$ to pure noise $\mathbf{x}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ at $t = 1$. A special case is known as the Variance-Preserving SDE (VP-SDE), where we can write the transition density $p_t(\mathbf{x}_t | \mathbf{x}_0)$ as

$$p(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad (3)$$

for $\bar{\alpha}_t = e^{\int_0^t \beta(s) ds}$ [20].

The reverse process aims to remove the noise from $\mathbf{x}_1$ and draw a sample from $\mathbf{x}_0$ following the reverse-time SDE

$$d\mathbf{x}_t = \left[-\frac{\beta(t)}{2}\mathbf{x}_t - \beta(t)\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) \right] dt + \sqrt{\beta(t)}d\bar{\mathbf{w}}_t, \quad (4)$$

where t and the new Wiener process $\bar{\mathbf{w}}_t$ both run in reverse. A seminal result of Anderson gives that this reverse process has the same probability distribution as the forward process for every t [1]. In order to estimate the time-dependent score function $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$, diffusion models use the denoising score matching technique to train a neural network $s_\theta(\cdot)$ that predicts the noise added to a datapoint, which approximates the score function [40]. Sampling from $p(\mathbf{x})$ is performed by solving a discrete-time version of Equation (4) over T steps, where $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ [20, 37].

2.2. Posterior Sampling

In inverse problems, we are given measurements $\mathbf{y}$ as in Equation (1), and the goal is to recover the underlying clean $\mathbf{x}_0$ given $\mathbf{y}$. Throughout the paper, we will refer to $\mathcal{A}$ as the known forward operator. Because $\mathcal{A}$ is not invertible in general, the inverse problem can be highly ill-posed, necessitating data modelling assumptions.

A popular Bayesian strategy to solve the problem is to develop samplers for $p(\mathbf{x} | \mathbf{y})$ assuming we have a prior for $p(\mathbf{x})$ modeled by a pretrained generative model such as a diffusion model. For instance we can extend Equation (4) to sample from $p(\mathbf{x} | \mathbf{y})$ by using a conditional score function

$$d\mathbf{x}_t = \left[-\frac{\beta(t)}{2}\mathbf{x}_t - \beta(t)\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t | \mathbf{y}) \right] dt + \sqrt{\beta(t)}d\bar{\mathbf{w}}_t. \quad (5)$$

While one could train a conditional diffusion model to approximate the conditional score, this would require retraining models for every inverse problem we wish to solve [33].

预训练无条件扩散模型中，第二项通常难以解析求解，因为 $\mathbf{x}_t$ 是复杂去噪过程的结果。因此在实际应用中，我们通常采用近似方法处理该项，具体做法是首先利用已学习的扩散模型计算 $\boldsymbol{\mu}_{0|t} = \mathbb{E}[\mathbf{x}_0|\mathbf{x}_t]$ （即Tweedie估计[17]），并沿损失函数的梯度进行优化，该损失函数用于衡量 $A(\boldsymbol{\mu}_{0|t})$ 与 $\mathbf{y}$ 之间的偏差。虽然直观合理，但Tweedie估计的近似误差难以控制。先前的研究尝试通过设计启发式方法来解决这一问题，确保对 $\mathbf{x}_t$ 的修改能保持扩散模型所学习的中间数据流形。然而，这些方法对数据流形的结构做出了强有力的假设，例如线性，或者要求 $\boldsymbol{\mu}_{0|t}$ 在整个迭代过程中保持在干净的数据流形上[10, 42]。

本研究旨在通过频域图像分析，更深入地理解和控制该近似误差。我们的主要贡献包括：

1. 我们开发了一种新的无监督扩散式图像复原算法，记为 **F**requency **G**uided **P**osterior **S**ampling (FGPS)。该算法利用了扩散模型在频域中分层生成图像的观察结果[30]。因此，为控制估计 $\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$ 时的近似误差，我们对 $\mathbf{y}$ 应用时变低通滤波器，该滤波器在反扩散过程中逐步包含更高频率范围。
2. 随后，我们对数据分布假设下线性逆问题后验采样方法的近似误差进行了精确的理论分析。接着，基于自然图像的频率分布特征，我们给出了精确的正向算子示例，其中FGPS的近似误差较先前研究更小。
3. 在实际应用中，我们为低通滤波器开发了一种数据依赖型课程，该课程使逆向过程能够适应数据分布的频谱特性。尽管实现简便，但我们的算法在具有挑战性的线性和非线性图像复原问题（如运动去模糊和图像去雾）上显著提升了性能。

2. 背景

2.1. 扩散模型

扩散模型是一类基于分数的生成模型，通过学习逆渐变噪声过程[20, 36]，从给定的数据分布中采样。该噪声过程被称为前向过程，在连续时间中表示为Ito随机微分方程。

(SDE) 对于 $t \in [0, 1]$

$$d\mathbf{x}_t = -\frac{1}{2}\beta(t)\mathbf{x}_t dt + \sqrt{\beta(t)}d\mathbf{w}_t. \quad (2)$$

该SDE的特征是其漂移系数 $-\frac{1}{2}\beta(t)$ 和扩散系数 $\sqrt{\beta(t)}$ ，其中 $\mathbf{w}_t$ 表示区间 $[0, 1]$ 上的标准维纳过程。该SDE的初始条件是一个数据点 $\mathbf{x}_0 \sim p(\mathbf{x})$ 。漂移和扩散系数的选择使得 $\mathbf{x}_t$ 从 $t=0$ 时的干净数据分布，到 $t=1$ 时的纯噪声 $\mathbf{x}_1 \sim N(\mathbf{0}, \mathbf{I})$ 。一个特殊情形被称为方差保持SDE (VP-SDE)，其中我们可以将转移密度 $p_t(\mathbf{x}_t|\mathbf{x}_0)$ 表示为

$$p(\mathbf{x}_t|\mathbf{x}_0) = N(\sqrt{\alpha_t}\mathbf{x}_0, (1-\alpha_t)\mathbf{I}), \quad (3)$$

为了 $\bar{\alpha}_t = e^{\int_0^t \beta(s) ds}$ [20].

反向过程旨在去除 $\mathbf{x}_1$ 中的噪声并从 $\mathbf{x}_0$ 中抽取样本，遵循逆时间SDE

$$d\mathbf{x}_t = \left[-\frac{\beta(t)}{2}\mathbf{x}_t - \beta(t)\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) \right] dt + \sqrt{\beta(t)}d\bar{\mathbf{w}}_t, \quad (4)$$

其中 t 和新的维纳过程 $\bar{\mathbf{w}}_t$ 两者均反向运行。Anderson的一个开创性结果表明，对于每个 t [1]，该反向过程的概率分布与正向过程相同。为了估计时间依赖的评分函数 $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ ，扩散模型采用去噪评分匹配技术训练神经网络 $s_\theta(\cdot)$ ，该网络预测数据点所添加的噪声，从而近似评分函数[40]。从 $p(\mathbf{x})$ 中采样是通过在 T 步长上求解方程(4)的离散时间版本来实现的，其中 $\mathbf{x}_T \sim N(\mathbf{0}, \mathbf{I})$ [20, 37]。

2.2. 后验抽样

在反问题中，我们给定测量值 $\mathbf{y}$ （如公式(1)所示），目标是根据 $\mathbf{y}$ 恢复底层的清洁 $\mathbf{x}_0$ 。本文中，我们将 A 称为已知的正向算子。由于 A 通常不可逆，该反问题可能高度不适定，因此需要数据建模假设。

解决该问题的一种流行贝叶斯策略是开发用于 $p(\mathbf{x}|\mathbf{y})$ 的采样器，前提是假设我们已通过预训练生成模型（如扩散模型）对 $p(\mathbf{x})$ 进行了先验建模。例如，我们可以将公式(4)扩展为通过使用条件评分函数从 $p(\mathbf{x}|\mathbf{y})$ 中采样。

$$d\mathbf{x}_t = \left[-\frac{\beta(t)}{2}\mathbf{x}_t - \beta(t)\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y}) \right] dt + \sqrt{\beta(t)}d\bar{\mathbf{w}}_t. \quad (5)$$

虽然可以训练条件扩散模型来近似条件评分，但这需要针对我们希望解决的每个逆问题重新训练模型[33]。

Instead, we use Bayes rule to write the conditional score as

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y}) = \underbrace{\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)}_{\text{unconditional score}} + \underbrace{\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)}_{\text{noisy likelihood score}} \quad (6)$$

While a pretrained diffusion model effectively approximates the unconditional score, the noisy likelihood score remains intractable due to its factorization

$$p(\mathbf{y} | \mathbf{x}_t) = \int p(\mathbf{y} | \mathbf{x}_0) p(\mathbf{x}_0 | \mathbf{x}_t) d\mathbf{x}_0, \quad (7)$$

where $p(\mathbf{x}_0 | \mathbf{x}_t)$ represents the complex distribution of denoised estimates of $\mathbf{x}_t$. Even though $p(\mathbf{x}_0 | \mathbf{x}_t)$ can be complex, the Tweedie’s estimate gives us the posterior mean $\boldsymbol{\mu}_{0|t} = \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]$ as a function of the unconditional score function [17] (refer to Appendix A for more background). Thus, some unsupervised methods approximate $p(\mathbf{x}_0 | \mathbf{x}_t)$ using $\boldsymbol{\mu}_{0|t}$, for example as a Dirac delta around $\boldsymbol{\mu}_{0|t}$ [9] or an isotropic Gaussian distribution with mean $\boldsymbol{\mu}_{0|t}$ [35]. This results in tractable approximations for the conditional score $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y})$, but introduces a potentially large approximation error. In this work, we propose a novel approximation for the conditional score and quantify its approximation gap exactly under distributional assumptions on $\mathbf{x}_0$.

2.3. Signal Processing Basics

Our method relies on the frequency decomposition of natural images, so in this section, we review the terminology used for the remainder of the paper. The Discrete Fourier Transform (DFT) of a finite signal $\mathbf{x} \in \mathbb{R}^n$, denoted as $\mathcal{F}(\mathbf{x})$, maps the signal to the frequency domain. At each DFT sample frequency $f_k = \frac{k}{n}$ for $k = 1, \dots, n-1$, the DFT value $\mathcal{F}(\mathbf{x})[f_k]$ represents the complex-valued frequency component in the signal. We define the signal’s *power spectral density* as the expected squared magnitude (the power) of the signal at different sample frequencies

$$S(f_k) = \mathbb{E}_{\mathbf{x}} \left[\frac{|\mathcal{F}(\mathbf{x})[f_k]|^2}{n} \right]. \quad (8)$$

When $S(f_k) = c|f_k|^{-\beta}$ for constants $c, \beta > 0$, we say that the signal follows a power law in the frequency domain with parameters c, β .

3. Related Work

Inverse problems of the form in Equation (1) have been solved for many years with different priors, such as sparse/low-rank priors [4–6] or generative priors [3, 29]. In this work, we focus on posterior sampling methods that use pretrained diffusion models and approximate the conditional score to perform guidance [8, 9, 19, 31, 35, 37, 41]. These have been applied to inverse problems such as image deblurring, super-resolution [11], and image inpainting

[9, 27]. However, they can incur significant approximation errors as we will illustrate in Section 5. Works such as [10, 42] propose methods to mitigate these approximation errors, but rely on strong assumptions on the data manifold. In this work, we use a more practical assumption that the data obeys a power law in the frequency domain [34] in order to develop a new approximation.

A line of work has studied diffusion models in a transformed space to perform guidance for inverse problems [23, 24]. This is in fact closely related to our work as our work is akin to performing diffusion in the Fourier domain instead, similar to [28]. Building on recent connections between diffusion models and spectral autoregression [15], our work differs from previous approaches [8, 37] by using a frequency-guided measurement schedule rather than one based on the unconditional diffusion model’s variance schedule. We refer the reader to [12] for a comprehensive survey on using diffusion models for inverse problems.

4. Frequency Guided Posterior Sampling

As illustrated in Section 2.2, the main challenge in developing unsupervised inverse problem solvers is estimating the noisy likelihood score $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$, which depends on the intractable denoising distribution $p(\mathbf{x}_0 | \mathbf{x}_t)$. It is common to substitute this denoising distribution with an approximation that depends only on the posterior mean $\boldsymbol{\mu}_{0|t} = \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]$, which is tractable using Tweedie’s formula. For example, Diffusion Posterior Sampling (DPS) approximates the noisy likelihood score with $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \boldsymbol{\mu}_{0|t})$ [9]. However, these methods can incur a significant approximation error by using simple distributions centered around the posterior mean instead of the true distribution.

The key challenge is ensuring that $\mathbf{x}_t$ stays on the data manifold learned by the diffusion model even after it is updated using an approximation of the gradient $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$. Prior approaches address this by assuming that the data manifold is linear or that $\boldsymbol{\mu}_{0|t}$ remains on the clean data manifold [10, 42]. However, these assumptions are often violated in practice. Our main insight is to use the observation that diffusion models generate images hierarchically. Specifically, recent work has connected diffusion models and inverse heat dissipation, concluding that diffusion models naturally generate images from low frequencies (corresponding to coarse features) to high frequencies (corresponding to fine-grained features) [30]¹. If there is a mismatch in frequency content between $\boldsymbol{\mu}_{0|t}$ and the measurement $\mathbf{y}$, updating $\mathbf{x}_t$ by following an approximation of $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$ can move $\mathbf{x}_t$ off the learned data manifold. This problem is exacerbated during the early stages of

¹This effect is because the forward process noise schedule removes high frequency components proportional to the variance of the noise.

相反，我们采用贝叶斯规则将条件评分表示为

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y}) = \underbrace{\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)}_{\text{无条件评分噪声似然评分}} + \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t) \quad (6)$$

尽管预训练扩散模型能有效近似无条件评分，但由于其分解特性，噪声似然评分仍难以处理。

$$p(\mathbf{y} | \mathbf{x}_t) = \int p(\mathbf{y} | \mathbf{x}_0) p(\mathbf{x}_0 | \mathbf{x}_t) d\mathbf{x}_0, \quad (7)$$

其中 $p(\mathbf{x}_0 | \mathbf{x}_t)$ 表示去噪估计值 $\mathbf{x}_t$ 的复分布。尽管 $p(\mathbf{x}_0 | \mathbf{x}_t)$ 可能是复数，但Tweedie估计量给出了后验均值 $\boldsymbol{\mu}_{0|t} = \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]$ 作为无条件评分函数[17]的函数（更多背景参见附录A）。因此，某些无监督方法通过 $\boldsymbol{\mu}_{0|t}$ 来近似 $p(\mathbf{x}_0 | \mathbf{x}_t)$ ，例如将其视为 $\boldsymbol{\mu}_{0|t}$ 的狄拉克 δ 函数[9]或均值为 $\boldsymbol{\mu}_{0|t}$ 的各向同性高斯分布[35]。这为条件得分 $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y})$ 提供了可处理的近似方法，但可能引入较大的近似误差。在本研究中，我们提出了一种新的条件得分近似方法，并在 $\mathbf{x}_0$ 的分布假设下精确量化了其近似差距。

2.3. 信号处理基础

我们的方法依赖于自然图像的频率分解，因此在本节中，我们将回顾本文后续使用的术语。有限信号 $\mathbf{x} \in \mathbb{R}^n$ 的离散傅里叶变换（DFT），记为 $F(\mathbf{x})$ ，将信号映射到频域。在每个DFT采样频率 $f_k = \frac{k}{n}$ for $k = 1, \dots, n-1$ ，DFT值 $F(\mathbf{x})[f_k]$ 表示信号中的复值频率分量。我们将信号的功率谱密度定义为信号在不同采样频率处的期望平方幅度（即功率）。

$$S(f_k) = \mathbb{E}_{\mathbf{x}} \left[\frac{|\mathcal{F}(\mathbf{x})[f_k]|^2}{n} \right]. \quad (8)$$

当 $S(f_k) = c|f_k|^{-\beta}$ 对于常数 $c, \beta > 0$ 时，我们说该信号在频域中遵循幂律，其参数为 c, β 。

3. 相关工作

方程（1）中形式的逆问题多年来已通过不同先验方法得到解决，例如稀疏/低秩先验[4–6]或生成先验[3, 29]。本研究聚焦于采用预训练扩散模型的后验采样方法，通过近似条件评分实现引导[8, 9, 19, 31, 35, 37, 41]。这些方法已应用于图像去模糊、超分辨率[11]及图像修复等逆问题。

[9, 27]。然而，正如我们将在第5节中说明的，它们可能会产生显著的近似误差。诸如[10, 42]等研究提出了缓解这些近似误差的方法，但依赖于对数据流形的强假设。在本工作中，我们采用了一个更实用的假设，即数据在频域中遵循幂律分布[34]，以开发一种新的近似方法。

有一类研究通过在变换空间中研究扩散模型来为逆问题提供引导[23, 24]。这实际上与我们的工作密切相关，因为我们的工作类似于在傅里叶域中进行扩散，类似于[28]。基于扩散模型与谱自回归之间最近的联系[15]，我们的工作与先前的方法[8, 37]的不同之处在于，我们使用的是频率引导的测量计划，而不是基于无条件扩散模型方差计划的测量计划。关于使用扩散模型解决逆问题的全面综述，我们建议读者参阅[12]。

4. 频率引导后验抽样

如第2.2节所示，开发无监督逆问题求解器的主要挑战在于估计带噪似然得分 $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$ ，该得分依赖于难以处理的去噪分布 $p(\mathbf{x}_0 | \mathbf{x}_t)$ 。通常仅依赖后验均值的近似值来替代该去噪分布 $\boldsymbol{\mu}_{0|t} = \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]$ ，该近似可通过Tweedie公式求解。例如，扩散后验采样（DPS）通过 $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \boldsymbol{\mu}_{0|t})$ [9] 来近似带噪似然得分。然而，这些方法可能因使用以后验均值为中心的简单分布而非真实分布而产生显著的近似误差。

关键挑战在于确保 $\mathbf{x}_t$ 即使在使用梯度近似 $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$ 更新后仍能保持在扩散模型学习到的数据流形上。先前方法通过假设数据流形是线性的或 $\boldsymbol{\mu}_{0|t}$ 保持在干净数据流形上来解决这一问题[10, 42]。然而这些假设在实践中常被违反。我们的主要洞见在于利用扩散模型生成图像的层次化特征这一观察结果。具体而言，近期研究将扩散模型与逆热耗散联系起来，得出扩散模型自然从低频（对应粗粒度特征）到高频（对应细粒度特征）生成图像的结论[30]¹。如果 $\boldsymbol{\mu}_{0|t}$ 与测量值 $\mathbf{y}$ 的频率成分存在不匹配，那么通过近似 $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$ 来更新 $\mathbf{x}_t$ ，可能会使 $\mathbf{x}_t$ 脱离已学习的数据流形。该问题在早期阶段会加剧。

¹ 这种效应源于前向过程噪声调度会消除与噪声方差成比例的高频分量。

the reverse process since the early stages play a crucial role in ensuring the generated output matches the fundamental low-frequency structure of the measurements.

We propose a simple fix to this issue by retaining only low frequency components of the measurements at the initial steps of the reverse process and progressively adding high frequency components in later steps. To implement this, we convolve the original measurement $\mathbf{y}$ with a time-varying low-pass filter $\mathbf{k}_t$. Denoting $\phi_t(\mathbf{y})$ as this convolution, the approximation we propose for the noisy likelihood score is

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t) \approx \nabla_{\mathbf{x}_t} \log p(\phi_t(\mathbf{y}) | \boldsymbol{\mu}_{0|t}), \quad (9)$$

where $\boldsymbol{\mu}_{0|t}$ denotes the Tweedie estimate $\mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]$ predicted by the diffusion model. In order to compute the above approximation, we need to compute the model likelihood for the new measurements. Since convolution of a signal $\mathbf{y}$ with a low pass filter can be represented as a circulant matrix multiplied by $\mathbf{y}$, we have that $\phi_t(\mathbf{y}) = \mathbf{C}_t \mathbf{y}$ where $\mathbf{C}_t$ is a time-dependent circulant matrix representing $\mathbf{k}_t$. This circulant matrix has its first row as the flattened $\mathbf{k}_t$ and all other rows as cyclic shifts. Thus, using properties of the multivariate normal distribution, we have that the updated model likelihood is

$$p(\phi_t(\mathbf{y}) | \mathbf{x}_0) = \mathcal{N}(\mathbf{C}_t \mathbf{A}(\mathbf{x}_0), \sigma_y^2 \mathbf{C}_t \mathbf{C}_t^T). \quad (10)$$

Given this likelihood, our approximation of the noisy likelihood score becomes

$$p(\phi_t(\mathbf{y}) | \boldsymbol{\mu}_{0|t}) = \mathcal{N}(\mathbf{C}_t \mathbf{A}(\boldsymbol{\mu}_{0|t}), \sigma_y^2 \mathbf{C}_t \mathbf{C}_t^T), \quad (11)$$

and its corresponding gradient is

$$\nabla_{\mathbf{x}_t} \log p(\phi_t(\mathbf{y}) | \boldsymbol{\mu}_{0|t}) = \mathbf{S}_t \nabla_{\mathbf{x}_t} \|\mathbf{C}_t \mathbf{y} - \mathbf{C}_t \mathbf{A}(\boldsymbol{\mu}_{0|t})\|_2^2, \quad (12)$$

where $\mathbf{S}_t = (\sigma_y^2 \mathbf{C}_t \mathbf{C}_t^T)^{-1}$. Since $\boldsymbol{\mu}_{0|t}$ can be computed using one network evaluation of the diffusion model, its gradient with respect to $\mathbf{x}_t$ can be computed via backpropagation, as in DPS. Alternating unconditional sampling and conditional sampling using their corresponding score functions gives us our algorithm, which we call **Frequency Guided Posterior Sampling (FGPS)** outlined in Algorithm 1.

Dataset-informed choice of $\mathbf{k}_t$. An important choice for our method is choosing the sequence $\{\mathbf{k}_t\}_{t=1}^T$. Each $\mathbf{k}_t$ is a low-pass filter that should allow frequencies up till a threshold τ_t and then strongly attenuate any frequencies above τ_t . In order to encode that the initial steps of the reverse process should only use low frequency components of $\mathbf{y}$, we require that τ_T should start off small and increase as t decreases. We denote the sequence of τ_t as our frequency curriculum. As datasets can have varying frequency characteristics, our

method gives flexibility to set the frequency curriculum in a data-dependent way, as explained in Section 6.

Efficient Implementation. We efficiently compute $\phi_t(\cdot)$ in the Fourier domain. Specifically, $\mathbf{C}_t$ can be diagonalized as $\mathbf{C}_t = \mathbf{F}^{-1} \mathbf{M} \mathbf{F}$, where $\mathbf{F}$ denotes the Discrete Fourier Transform matrix [39]. Using this property along with the fact that $\mathbf{F}$ is an orthogonal matrix with determinant 1, we have that $p(\phi_t(\mathbf{y}) | \boldsymbol{\mu}_{0|t}) = p(\mathbf{F} \phi_t(\mathbf{y}) | \boldsymbol{\mu}_{0|t}) = p(\mathbf{M} \mathbf{F} \mathbf{y} | \boldsymbol{\mu}_{0|t})$. Thus, we bypass storing the large circulant matrix and compute $\phi_t(\mathbf{y})$ and $\phi_t(\mathbf{A}(\boldsymbol{\mu}_{0|t}))$ by first taking the Fast Fourier Transform of the input and applying a binary mask in the frequency domain, which adds minimal computational overhead (see Appendix E).

Algorithm 1 Frequency Guided Posterior Sampling

Require: $\mathbf{y}, \mathcal{A}, \{\phi_t(\cdot), \mathbf{S}_t, \alpha_t, \tilde{\sigma}_t\}_{t=1}^T$

- 1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 2: **for** $t \leftarrow T$ **to** 1 **do**
- 3: $\hat{\mathbf{s}} \leftarrow s_\theta(\mathbf{x}_t, t)$ // Predict $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$
- 4: $\bar{\alpha}_t \leftarrow \prod_{j=1}^t \alpha_j$
- 5: $\boldsymbol{\mu}_{0|t} \leftarrow \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t + (1 - \bar{\alpha}_t)\hat{\mathbf{s}})$ // Tweedie estimate
- 6: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 7: $\mathbf{x}'_{t-1} \leftarrow \frac{\sqrt{\alpha_t(1-\bar{\alpha}_{t-1})}}{1-\bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}(1-\alpha_t)}}{1-\bar{\alpha}_t} \boldsymbol{\mu}_{0|t} + \tilde{\sigma}_t \mathbf{z}$ // DDIM Sampling
- 8: $\mathbf{x}_{t-1} \leftarrow \mathbf{x}'_{t-1} - \mathbf{S}_t \nabla_{\mathbf{x}_t} \|\phi_t(\mathbf{y}) - \phi_t(\mathbf{A}(\boldsymbol{\mu}_{0|t}))\|_2^2$ // FGPS Approximation of $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$
- 9: **end for**

5. Theoretical Analysis

In this section, we utilize distributional assumptions to provide a precise theoretical analysis of the approximation errors induced by traditional posterior sampling methods as well as our proposed method.

5.1. Mind the (Approximation) Gap

Since we rely on a spectral characterization of the reverse diffusion process, our analysis requires two key properties.

1. We need to analyze the frequency characteristics of the data as well as the interplay between the frequency characteristics of the data and the forward operator.
2. To quantify the approximation gap, we need to compute the true conditional score.

For the first property, we look at the power spectral density of the data, as defined in Equation (8), which tells us how much the signal's power is distributed across different frequencies. For the second property, it is natural to suppose the clean data $\mathbf{x}_0$ comes from a multivariate Gaussian distribution. Although these two properties seem conflicting, the following lemma establishes an equivalence between the data power spectral density and the data covariance.

逆向过程自早期阶段起即发挥关键作用，以确保生成的输出与测量的基本低频结构相匹配。

我们提出一个简单的解决方案：在逆向过程的初始步骤中仅保留测量的低频分量，并在后续步骤中逐步加入高频分量。为实现这一点，我们将原始测量 $\mathbf{y}$ 与时变低通滤波器 $\mathbf{k}_t$ 进行卷积。将 $\phi_t(\mathbf{y})$ 表示为该卷积运算，我们提出的噪声似然分数近似方法为

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t) \approx \nabla_{\mathbf{x}_t} \log p(\phi_t(\mathbf{y}) | \boldsymbol{\mu}_{0|t}), \quad (9)$$

其中 $\boldsymbol{\mu}_{0|t}$ 表示扩散模型预测的Tweedie估计量 $E[\mathbf{x}_0 | \mathbf{x}_t]$ 。为了计算上述近似值，我们需要计算新测量值的模型似然。由于信号 $\mathbf{y}$ 与低通滤波器的卷积可表示为循环矩阵乘以 $\mathbf{y}$ ，因此有 $\phi_t(\mathbf{y}) = \mathbf{C}_t \mathbf{y}$ ，其中 $\mathbf{C}_t$ 是表示 $\mathbf{k}_t$ 的时变循环矩阵。该循环矩阵的第一行是展平的 $\mathbf{k}_t$ ，其余各行均为循环移位。因此，利用多元正态分布的性质，我们得到更新后的模型似然为

$$p(\phi_t(\mathbf{y}) | \mathbf{x}_0) = N(\mathbf{C}_t \mathbf{A}(\mathbf{x}_0), \sigma_{2y} \mathbf{C}_t \mathbf{C}_t^T). \quad (10)$$

基于此可能性，我们对噪声似然评分的近似计算结果为

$$p(\phi_t(\mathbf{y}) | \boldsymbol{\mu}_{0|t}) = N(\mathbf{C}_t \mathbf{A}(\boldsymbol{\mu}_{0|t}), \sigma_{2y} \mathbf{C}_t \mathbf{C}_t^T), \quad (11)$$

及其对应梯度为

$$\nabla_{\mathbf{x}_t} \log p(\phi_t(\mathbf{y}) | \boldsymbol{\mu}_{0|t}) = \mathbf{S}_t \nabla_{\mathbf{x}_t} \|\mathbf{C}_t \mathbf{y} - \mathbf{C}_t \mathbf{A}(\boldsymbol{\mu}_{0|t})\|_2^2, \quad (12)$$

其中 $\mathbf{S}_t = (\sigma_{2y} \mathbf{C}_t \mathbf{C}_t^T)^{-1}$ 。由于 $\boldsymbol{\mu}_{0|t}$ 可通过一次扩散模型的网络评估计算得出，其关于 $\mathbf{x}_t$ 的梯度可通过反向传播计算，如DPS中所述。交替使用无条件采样和条件采样及其对应的评分函数，我们得到了算法，称为**F**频率**G**引导**P**后验**S**采样（FGPS），如算法1所述。

数据集驱动的 $\mathbf{k}_t$ 选择。我们方法的一个重要选择是选择序列 $\{\mathbf{k}_t\}_{t=1}^T$ 。Each $\mathbf{k}_t$ is a 低通滤波器应允许频率达到阈值 $=_t$ ，然后强烈衰减高于 $=_t$ 的频率。为了编码逆过程的初始步骤仅使用 $\mathbf{y}$ 的低频分量，我们要求 $=_T$ 应从小开始并随 t 减小而增加。我们将 $=_t$ 的序列称为频率课程。由于数据集可能具有不同的频率特性，我们的

如第6节所述，该方法可灵活地以数据依赖的方式设置频率课程。

高效实现。我们在傅里叶域中高效计算 $\phi_t(\cdot)$ 。具体而言， $\mathbf{C}_t$ 可对角化为 $\mathbf{C}_t = \mathbf{F}^{-1} \mathbf{M} \mathbf{F}$ ，其中 $\mathbf{F}$ 表示离散傅里叶变换矩阵[39]。利用这一性质以及 $\mathbf{F}$ 是行列式为1的正交矩阵这一事实，我们得到 $p(\phi_t(\mathbf{y}) | \boldsymbol{\mu}_{0|t}) = p(\mathbf{F} \phi_t(\mathbf{y}) | \mathbf{F} \boldsymbol{\mu}_{0|t}) = p(\mathbf{M} \mathbf{F} \mathbf{y} | \mathbf{F} \boldsymbol{\mu}_{0|t})$ 。因此，我们跳过存储大型循环矩阵的步骤，通过首先对输入进行快速傅里叶变换并在频域应用二进制掩码，来计算 $\phi_t(\mathbf{y})$ 和 $\phi_t(\mathbf{A}(\boldsymbol{\mu}_{0|t}))$ ，这种方法仅增加极小的计算开销（参见附录E）。

算法1 频率引导后验采样

要求 : $\mathbf{y}, \mathbf{A}, \{\phi_t(\cdot)\}, \mathbf{S}_t, \sigma_t, \tilde{\sigma}_t\}_{t=1}^T$
1: $\mathbf{x}_T \sim N(\mathbf{0}, \mathbf{I})$
2: **对于** $t \leftarrow T$ **至** 1 **执行**
3: $\hat{\mathbf{s}} \leftarrow s_\theta(\mathbf{x}_t, t)$ // 预测 $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$
4: $\bar{\alpha}_t \leftarrow \prod_{j=1}^t \alpha_j$
5: $\boldsymbol{\mu}_{0|t} \leftarrow \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t + (1 - \bar{\alpha}_t)\hat{\mathbf{s}})$ // Tweedie估计
6: $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$
7: $\mathbf{x}'_{t-1} \leftarrow \frac{\sqrt{\alpha_t(1-\bar{\alpha}_{t-1})}}{1-\bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}(1-\alpha_t)}}{1-\bar{\alpha}_t} \boldsymbol{\mu}_{0|t} + \tilde{\sigma}_t \mathbf{z}$
 // DDIM 抽样
8: $\mathbf{x}_{t-1} \leftarrow \mathbf{x}'_{t-1} - \mathbf{S}_t \nabla_{\mathbf{x}_t} \phi_t(\mathbf{y}) - \mathbf{P}_t(\mathbf{A}(\boldsymbol{\mu}_{0|t}))$ // FGPS 近似 $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$
9: **端**

5. 理论分析

在本节中，我们利用分布假设对传统后验抽样方法及我们提出的方法所导致的近似误差进行了精确的理论分析。

5.1. 注意（近似）差距

由于我们依赖于反向扩散过程的光谱表征，因此我们的分析需要两个关键特性。

1. 需要分析数据的频率特性以及数据频率特性与前向算子之间的相互作用。
2. 为量化近似误差，需计算真实的条件评分。

对于第一个特性，我们关注数据的功率谱密度（如公式8所定义），它揭示了信号功率在不同频率上的分布情况。第二个特性则基于一个自然假设：原始数据 $\mathbf{x}_0$ 服从多元高斯分布。尽管这两个特性看似矛盾，但下述引理证明了数据功率谱密度与数据协方差之间存在等价关系。

Lemma 1. Let $f_k = \frac{k}{n}$ for $k = 0, \dots, n-1$ denote the DFT sample frequencies for a signal of length n . Then, there exists a covariance matrix Σ_f such that for $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \Sigma_f)$, the following two properties hold. First, the signal $\mathbf{x}$ follows a power law in the frequency domain with parameters $c, \beta > 0$, i.e., it has a power spectral density $S(f_k) = c|f_k|^{-\beta}$ for the non-zero DFT sample frequencies f_k . Second, the eigenvalues of Σ_f are precisely $c|f_k|^{-\beta}$.

This is a special case of a classical result known as the Wiener-Khinchin Theorem [7]. Note that this does not imply that all random vectors that follow a power law in the frequency domain are Gaussian. For example, natural image data tends to follow a radially averaged power law in the frequency domain, but is far from Gaussian [34]. Nevertheless, this lemma gives a distributional assumption that mimics the spectral properties of natural image data, which we will show already elucidates failure cases of existing methods. Next, we use these assumptions to exactly quantify the approximation gap of our method.

Theorem 2. Suppose $\mathbf{x}_0$ is drawn from $\mathcal{N}(\mathbf{0}, \Sigma)$ and we are given linear measurements $\mathbf{y} = \mathbf{A}(\mathbf{x}_0) + \mathbf{z}$, where $\mathbf{A}(\mathbf{x}_0) = \mathbf{A}\mathbf{x}_0$ and $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma_y^2 \mathbf{I})$. Suppose that the intermediate value $\mathbf{x}_t$ of the continuous-time reverse diffusion from Equation (5) can be written as $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$ where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Then, we have that the true noisy likelihood score $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$ is

$$(\mathbf{A}\Gamma_t)^T (\mathbf{A}\Sigma_{0|t}\mathbf{A}^T + \sigma_y^2 \mathbf{I})^{-1} (\mathbf{y} - \mathbf{A}\mu_{0|t}), \quad (13)$$

where $\Gamma_t = \sqrt{\bar{\alpha}_t} \Sigma (\bar{\alpha}_t \Sigma + (1 - \bar{\alpha}_t) \mathbf{I})^{-1}$, $\mu_{0|t} = \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t] = \Gamma_t \mathbf{x}_t$ and $\Sigma_{0|t} = \text{Cov}[\mathbf{x}_0 | \mathbf{x}_t] = \Sigma - \sqrt{\bar{\alpha}_t} \Gamma_t \Sigma$. Moreover, the FGPS approximation $\nabla_{\mathbf{x}_t} \log p(\mathbf{C}_t \mathbf{y} | \mu_{0|t})$ can be analytically calculated as

$$(\mathbf{C}_t \mathbf{A}\Gamma_t)^T (\sigma_y^2 \mathbf{C}_t \mathbf{C}_t^T)^{-1} (\mathbf{C}_t \mathbf{y} - \mathbf{C}_t \mathbf{A}\mu_{0|t}). \quad (14)$$

The proof is in Appendix B. The difference of Equation (13) and Equation (14) gives a precise expression for the approximation gap of our method. The $\bar{\alpha}_t$ corresponds to the variance schedule of standard diffusion models such as DDPM [20]. Our assumption that $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$ isolates the effects of the approximation gap as the generated sample is exactly $\mathbf{x}_0$. Our framework also allows us to analytically study the approximation gap for other inverse problem solvers. Below, we give one example for the popular Diffusion Posterior Sampling (DPS) solver.

Corollary 3. Under the same assumptions and notation as in Theorem 2, we have that the DPS approximation $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mu_{0|t})$ can be analytically calculated as

$$\sigma_y^{-2} (\mathbf{A}\Gamma_t)^T (\mathbf{y} - \mathbf{A}\mu_{0|t}). \quad (15)$$

To the best of our knowledge, this is the first time any distributional assumptions have been used to characterize the exact approximation gap of any inverse problem solver. While [9] and [42] provide upper and lower bounds on this gap, these bounds do not exploit the structure of the data distribution and as such, can be loose. We formally compare the approximation gap of FGPS and DPS in Appendix C.

5.2. Approximation Gap on Synthetic Data

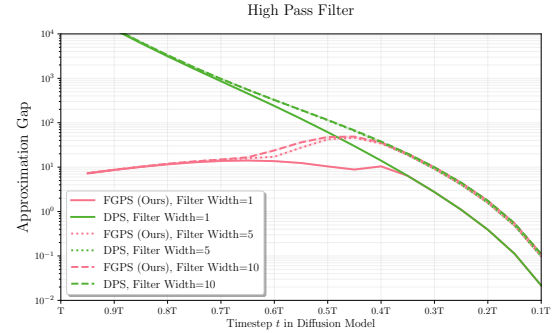


Figure 1. We show the exact approximation gap for FGPS and DPS across diffusion timesteps when the data follows a power law in the frequency domain and the forward operator is a high-pass filter (Dirac kernel minus a Gaussian kernel) of varying width. The width denotes the σ of the Gaussian kernel. Note the y -axis is in log-scale.

In this subsection, we probe the approximation gap of our method and DPS on synthetic data, and we give examples of forward operators where FGPS results in a significantly less approximation gap than DPS. In order to construct these examples, we note that Theorem 2 depends crucially on how the forward operator $\mathbf{A}$ interacts with the data covariance Σ , particularly through the term $\mathbf{A}\Sigma_{0|t}\mathbf{A}^T$. Intuitively, if the data covariance is chosen to be Σ_f from Lemma 1, the frequency characteristics of the data and the forward operator would be misaligned when the forward operator is a high-pass filter, so the measurement contains primarily high-frequency components. Some natural imaging systems fall into this regime, e.g., phase contrast microscopy where intensity measurements are high-pass filtered versions of refractive index.

To verify the above intuition, we begin by simulating 1-D signals $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \Sigma_f)$ that follow a power law in the frequency domain, which uses the construction of Σ_f given in Lemma 1. Then, the corresponding measurements for these signals are $\mathbf{y} = \mathbf{A}\mathbf{x}_0 + \mathbf{z}$ with $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, where $\mathbf{A}$ is the convolution matrix corresponding to convolution with a Dirac kernel minus a Gaussian kernel of standard deviation σ . This kernel approximates a high-pass filter.

Figure 1 shows the approximation gap of DPS and FGPS as a function of the diffusion timesteps (where higher

引理 1. 设 $f_k = \frac{k}{n}$ for $k = 0, \dots, n-1$ 表示长度为 n 的信号 $\mathbf{x}$ 的 DFT 样本频率。那么，存在一个协方差矩阵 $\mathbf{H}_f$ ，使得对于 $\mathbf{x} \sim N(\mathbf{0}, \mathbf{H}_f)$ ，以下两个性质成立。首先，信号 $\mathbf{x}$ 在频域中遵循幂律分布，参数为 $c, \gamma > 0$ ，即对于非零 DFT 样本频率 f_k ，其功率谱密度为 $S(f_k) = c|f_k|^{-\gamma}$ 。其次， $\mathbf{H}_f$ 的特征值恰好为 $c|f_k|^{-\gamma}$ 。

这是经典维纳-金钦定理[7]的一个特例。需要说明的是，这并不意味着所有在频域遵循幂律分布的随机向量都具有高斯特性。例如，自然图像数据在频域往往遵循径向平均幂律分布，但其分布形态与高斯分布存在显著差异[34]。尽管如此，该引理提出的分布假设仍能模拟自然图像数据的频谱特性，我们后续将通过实例证明，这种假设已能有效揭示现有方法的局限性。接下来，我们将基于这些假设对本方法的近似误差进行精确量化。

定理 2. 假设 $\mathbf{x}_0$ 从 $N(\mathbf{0}, \mathbf{H})$ 中抽取，且给定线性测量 $\mathbf{y} = A(\mathbf{x}_0) + \mathbf{z}$ ，其中 $A(\mathbf{x}_0) = \mathbf{A}\mathbf{x}_0$ 且 $\mathbf{z} \sim N(\mathbf{0}, \sigma_y^2 \mathbf{I})$ 。假设方程(5)中连续时间反向扩散的中间值 $\mathbf{x}_t$ 可表示为 $\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \mathbf{z}_t$ ，其中 $\mathbf{z}_t \sim N(\mathbf{0}, \mathbf{I})$ 。那么，我们得到真实的带噪似然得分 $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$ 是

$$(\mathbf{A}'_t)^T (\mathbf{A} \mathbf{H}_{0|t} \mathbf{A}^T + \sigma_y^2 \mathbf{I})^{-1} (\mathbf{y} - \mathbf{A}'_{0|t}), \quad (13)$$

其中 $\mathbf{A}'_t = \sqrt{1 - \alpha_t} \mathbf{H} (\sqrt{1 - \alpha_t} \mathbf{H} + (1 - \sqrt{1 - \alpha_t}) \mathbf{I})^{-1}$ ， $\mathbf{z}_{0|t} = E[\mathbf{x}_0 | \mathbf{x}_t] = \sqrt{1 - \alpha_t} \mathbf{x}_t$ 且 $\mathbf{H}_{0|t} = \text{Cov}[\mathbf{x}_0 | \mathbf{x}_t] = \mathbf{H} - \sqrt{1 - \alpha_t} \mathbf{H} \mathbf{H}'_t \mathbf{H}$ 。此外，FGPS 近似式 $\nabla_{\mathbf{x}_t} \log p(\mathbf{C}_t \mathbf{y} | \mathbf{z}_{0|t})$ 可通过解析计算得出

$$(\mathbf{C}_t \mathbf{A}'_t)^T (\sigma_y^2 \mathbf{C}_t \mathbf{C}_t^T)^{-1} (\mathbf{C}_t \mathbf{y} - \mathbf{C}_t \mathbf{A}'_{0|t}). \quad (14)$$

证明见附录B。方程(13)与方程(14)的差异给出了我们方法的近似间隙精确表达式。 α_t 对应于标准扩散模型（如 DDPM [20]）的方差分布。我们假设 $\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \mathbf{z}_t$ ，该假设使近似间隙的影响被隔离，因为生成的样本恰好为 $\mathbf{x}_0$ 。我们的框架还允许我们对其他反问题求解器的近似间隙进行解析研究。下文将给出一个流行的扩散后验采样（DPS）求解器的示例。

推论 3. 在与定理 2 相同的假设和符号下，我们有：DPS 近似 $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{z}_{0|t})$ 可以通过解析计算得出

$$\sqrt{1 - \alpha_t} \mathbf{y} - 2 (\mathbf{A}'_t)^T (\mathbf{y} - \mathbf{A}'_{0|t}). \quad (15)$$

据我们所知，这是首次利用分布假设来刻画任何逆问题求解器的精确近似差距。虽然文献[9]和[42]分别给出了该差距的上下界，但这些界值未利用数据分布的结构特性，因此可能不够精确。我们在附录C中对 FGPS 和 DPS 的近似差距进行了正式比较。

5.2. 合成数据上的近似间隙

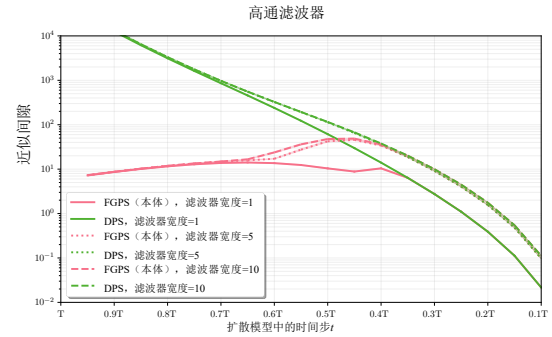


图1.我们展示了当数据在频域遵循幂律且前向算子为可变宽度的高通滤波器（狄拉克核减去高斯核）时，FGPS 和 DPS 在扩散时间步长上的精确近似间隙。宽度表示高斯核的 σ 。注意 y 轴采用对数刻度。

在本小节中，我们通过合成数据探究了本方法与 DPS 的近似间隙，并给出了前向算子的示例，其中 FGPS 结果的近似间隙显著小于 DPS。为构建这些示例，我们注意到定理 2 关键依赖于前向算子 $\mathbf{A}$ 与数据协方差 $\mathbf{H}$ 的交互方式，特别是通过项 $\mathbf{A} \mathbf{H}_{0|t} \mathbf{A}^T$ 。直观而言，若根据引理 1 将数据协方差设为 $\mathbf{H}_f$ ，当前向算子为高通滤波器时，数据与前向算子的频率特性将发生错位，导致测量结果主要包含高频分量。某些自然成像系统即属于这种情况，例如相位对比显微镜中强度测量是折射率的高通滤波版本。

为验证上述直觉，我们首先模拟 1 维信号 $\mathbf{x}_0 \sim N(\mathbf{0}, \Delta_f)$ ，这些信号在频域遵循幂律分布，其构造基于引理 1 中给出的 Δ_f 。随后，这些信号的对应测量结果为 $\mathbf{y} = \mathbf{A}\mathbf{x}_0 + \mathbf{z}$ ，其中 $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$ ，且 $\mathbf{A}$ 是对应于狄拉克核减去标准差为 σ 的高斯核卷积的卷积矩阵。该核近似为高通滤波器。

图 1 展示了 DPS 与 FGPS 的近似误差随扩散时间步长变化的函数关系（其中较高

timesteps correspond to a lower value of $\bar{\alpha}_t$) for the variance schedule used in the DDPM model for $T = 1000$ [20]. In accordance with our intuition, we see that the DPS approximation worsens significantly as we increase t , likely because higher frequency noise components in $\mathbf{x}_t$ are amplified by the high-pass filter. In contrast, we find our method has an approximation gap that is several orders of magnitude smaller than DPS in the early stages of the reverse process. As the early stages of the reverse process are when the generated image develops coarse features, we argue it is crucial that the approximation gap is small in these phases. Refer to Appendix E for the full experimental details.

6. Experiments

Although our theoretical results were restricted to cases where FGPS provably improved over existing methods, we now turn to empirical validation of FGPS on several real-world datasets and image restoration tasks.

6.1. Experimental Setup

Datasets. We evaluate our method on two datasets: the Flickr-Faces-HQ (FFHQ 256x256) dataset of high-resolution faces [22], and the ImageNet-1000 dataset of diverse objects [13]. We utilize pretrained diffusion models from [9] and [14] for the FFHQ and ImageNet datasets respectively, using a class-conditional model for ImageNet.

Tasks. We consider four challenging image restoration tasks for our experimental evaluation. The first three are linear inverse problems: Gaussian deblurring, motion deblurring, and deconvolution with a high-pass filter. The last is a nonlinear inverse problem: image dehazing. The details of the four forward operators we consider can be found in Appendix E. All measurements have Gaussian measurement noise added with standard deviation 0.05.

Baselines and Metrics. We compare against several competitive unsupervised methods. The first two baselines are popular unsupervised diffusion-based solvers: Diffusion Posterior Sampling (DPS) [9], a method that approximates the conditional score with a Dirac delta around the posterior mean, and Score-SDE/ILVR [8, 37], a method that uses a sequence of noisy measurements computed using the diffusion model variance schedule. Then, we compare to two state-of-the-art methods focused on mitigating the approximation gap problem: Manifold Constrained Gradient (MCG) [10] and Diffusion with Spherical Guidance (DSG) [42]. For image dehazing, we further compare to two problem-specific baselines: DoubleDIP [18] and AOD-Net [26], a supervised method trained to dehaze images from pairs of clean-hazy images. We use the same pretrained score function for all diffusion-based methods. We evaluate our method on two perceptual metrics, FID and LPIPS distance, and two distortion-based metrics, PSNR and SSIM.

Frequency Curriculum. For both FFHQ and ImageNet across all tasks, in our method we set τ_1 and τ_T to be $\frac{75 \cdot 100}{256}\%$ and $\frac{10 \cdot 100}{256}\%$ of the overall frequency range respectively. We set different frequency curricula for the two datasets as described in Section 4. We find the exponential schedule works well for face datasets with many high-frequency details, while the uniform schedule is better for diverse, multiclass datasets like ImageNet. We emphasize that $\mathbf{k}_t$ does not require extensive tuning, as a simple schedule already beats baseline methods (see Appendix G).

Step Size Schedule. In practice, we take $\mathbf{S}_t$ to be a hyperparameter as opposed to being fixed as $(\sigma_y^2 \mathbf{C}_t \mathbf{C}_t^T)^{-1}$ in order to further control the effect of the approximation error. We adopt the choice that DPS makes where the step size is $\frac{\kappa}{\|\phi_t(\mathbf{y}) - \phi(\mathcal{A}(\mu_{0|t}))\|_2} \mathbf{I}$ for a hyperparameter κ . Using the intuition from Figure 1 that the approximation gap is small in the initial steps of the reverse process, we set κ large initially and decay it following a cosine curve.

6.2. Results

Empirical Validation of Theoretical Results. In Figure 2, we consider two forward operators on an example from the FFHQ dataset: a) a standard Gaussian blurring low-pass filter, and b) the high-pass filter we studied in our theoretical analysis. As expected, prior approaches such as Diffusion Posterior Sampling [9] and variants [10, 42] perform well at Gaussian deblurring. However, when the forward operator is a high-pass filter, remarkably these approaches all fail to produce coherent outputs, which aligns with our theoretical results about the large approximation gap of existing methods. FGPS on the other hand is still able to produce high-quality reconstructions.

Results on Linear Inverse Problems. Table 1 shows our main results for linear inverse problems. On the FFHQ dataset, for simple tasks such as Gaussian deblurring, FGPS gives comparable performance to baselines, which already have strong reconstruction performance. However, on more complex image restoration tasks such as motion deblurring, our method outperforms the baselines in almost all metrics. Notably, even though Score-SDE/ILVR uses a sequence of noisy measurements, this sequence is not adapted to the frequency characteristics of the data, which results in lower quality reconstructions as compared to our method (see Appendix F for further discussion). In Figure 3, we see the qualitative differences between the methods. For high-pass filter operators, existing methods generate visually implausible images, indicating that the approximation errors push the denoised outputs away from the natural image manifold. In contrast, our method is able to accurately reconstruct the underlying ground truth image. For blurry images, we see that DPS tends to produce smoothed images, losing high-frequency details. While DSG is able to improve upon this

时间步长对应于 DDPM 模型中使用的方差调度 β_t 的较低值，其中 $T = 1000$ [20]。与我们的直觉一致，我们发现随着 t 的增加，DPS 近似显著恶化，这可能是由于 $\mathbf{x}_t$ 中的高频噪声成分被高通滤波器放大所致。相比之下，我们发现我们的方法在逆向过程的早期阶段具有比 DPS 小几个数量级的近似差距。由于逆向过程的早期阶段是生成图像形成粗略特征的阶段，我们认为这些阶段的近似差距必须很小。完整实验细节参见附录 E。

6. 实验

尽管我们的理论结果仅适用于 FGPS 能显著优于现有方法的案例，但我们现在将转向在多个真实数据集和图像修复任务上验证 FGPS 的实证效果。

6.1. 实验设置

数据集。 我们在两个数据集上评估了该方法：

高分辨率人脸数据集 Flickr-Faces-HQ (FFHQ 256x256) [22]，以及多样化物体数据集 ImageNet-1000 [13]。我们分别使用来自 [9] 和 [14] 的预训练扩散模型处理 FFHQ 和 ImageNet 数据集，并为 ImageNet 采用类别条件模型。

任务。 我们为实验评估设计了四个具有挑战性的图像修复任务。前三个属于线性逆问题：高斯去模糊、运动去模糊和高通滤波反卷积。最后一个属于非线性逆问题：图像去雾。我们考虑的四个前向算子的具体细节可参见附录 E。所有测量数据均添加了标准差为 0.05 的高斯测量噪声。

基线与指标。 我们与几种竞争性的无监督方法进行对比。前两个基线是流行的无监督扩散求解器：扩散后验采样 (DPS) [9]，该方法通过狄拉克 δ 函数在后验均值附近近似条件分数；以及 Score-SDE / ILVR [8, 37]，该方法利用扩散模型方差调度计算的一系列噪声测量值。随后，我们对比了两种专注于缓解近似差距问题的前沿方法：流形约束梯度 (MCG) [10] 和球面引导扩散 (DSG) [42]。针对图像去雾，我们进一步对比了两个问题特定基线：DoubleDIP [18] 和 AOD-Net [26]，后者是一种通过训练从干净-雾图像对中去雾的监督方法。所有基于扩散的方法均使用相同的预训练分数函数。我们采用两个感知指标 (FID 和 LPIPS 距离) 和两个基于失真的指标 (PSNR 和 SSIM) 来评估我们的方法。

频率课程。 对于 FFHQ 和 ImageNet 的所有任务，我们的方法中将 β_1 和 β_T 设置为 $\frac{75 \times 100\%}{256}$ and $\frac{10 \times 100\%}{256}$ 分别占整体频率范围的百分比。我们为两个数据集设置了不同的频率方案，具体如第 4 节所述。我们发现指数式调度方案在包含大量高频细节的面部数据集上表现良好，而均匀调度方案则更适合 ImageNet 这类多样化的多类别数据集。需要强调的是， $\mathbf{k}_t$ 方案无需大量调优，因为简单的调度方案就已经优于基线方法（详见附录 G）。

步长调度。 实际上，我们将 $\mathbf{S}_t$ 视为超参数而非固定值 $(\frac{1}{2} \gamma \mathbf{C}_t \mathbf{C}_t)^{-1}$ ，以便进一步控制近似误差的影响。我们采用 DPS 的设定方式，其中步长为 $\mathbf{k}_t(y) \mathbf{P}_t \cdot \frac{1}{\sqrt{A(\gamma, t) \gamma}} \mathbf{k}_2 \mathbf{I}$ ，其中 h 为超参数。基于图 1 的直观理解——反向过程初始阶段的近似间隙较小，我们初始时将 h 设为较大值，并使其按余弦曲线衰减。

6.2. 结果

理论结果的实实验验证。 在图 2 中，我们以 FFHQ 数据集中的一个示例考虑了两种前向算子：a) 标准高斯模糊低通滤波器，以及 b) 我们在理论分析中研究的高通滤波器。正如预期的那样，先前的方法如扩散后验采样 [9] 及其变体 [10, 42] 在高斯去模糊方面表现良好。然而，当前向算子为高通滤波器时，值得注意的是这些方法均未能产生连贯的输出，这与我们关于现有方法存在较大近似差距的理论结果一致。另一方面，FGPS 仍能生成高质量的重建结果。

线性逆问题研究结果。 表 1 展示了我们在线性逆问题上的主要成果。在 FFHQ 数据集上，对于高斯去模糊等简单任务，FGPS 方法的表现与已具备强重建性能的基线方法相当。然而在运动去模糊等更复杂的图像修复任务中，我们的方法在几乎所有指标上都优于基线方法。值得注意的是，尽管 Score-SDE / ILVR 采用噪声测量序列，但该序列未适配数据的频率特性，导致重建质量低于我们的方法（详见附录 F）。图 3 直观呈现了两种方法的差异：对于高通滤波算子，现有方法生成的图像缺乏视觉可信度，表明近似误差使去噪输出偏离自然图像流形；而我们的方法能精准还原原始真实图像。对于模糊图像，DPS 方法倾向于生成平滑化图像，导致高频细节丢失。虽然 DSG 能够改进这一点

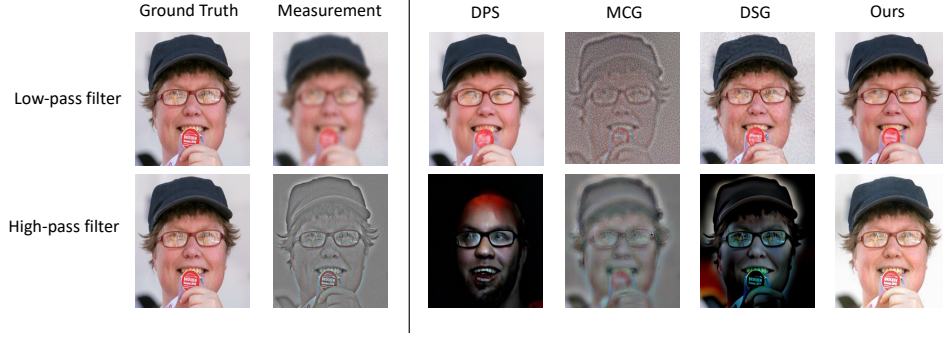


Figure 2. Even for linear inverse problems, SOTA diffusion-based methods (DPS, MCG, DSG) can struggle to solve inverse problems when the forward operator is convolution with a high-pass filter. In contrast, our method is still able to achieve a high quality reconstruction.

Method	Gaussian Blur				Motion Blur				High-Pass Filter			
	FID↓	LPIPS↓	PSNR↑	SSIM↑	FID↓	LPIPS↓	PSNR↑	SSIM↑	FID↓	LPIPS↓	PSNR↑	SSIM↑
FFHQ Dataset												
Score-SDE/ILVR [8, 37]	58.79	0.266	23.34	0.648	69.10	0.387	18.59	0.457	85.35	0.356	11.86	0.600
MCG [10]	310.33	1.035	11.64	0.143	296.93	0.771	11.54	0.127	151.77	0.581	<u>12.48</u>	0.466
DPS [9]	21.69	0.116	26.01	0.743	<u>27.15</u>	<u>0.153</u>	24.17	<u>0.688</u>	121.57	0.694	6.17	0.222
DSG [42]	44.63	0.294	22.75	0.633	32.74	0.257	25.66	0.687	<u>17.81</u>	<u>0.246</u>	12.12	<u>0.695</u>
FGPS (Ours)	<u>21.89</u>	<u>0.139</u>	<u>25.44</u>	<u>0.727</u>	20.48	0.127	<u>25.48</u>	0.723	10.02	0.124	18.79	0.739
ImageNet Dataset												
Score-SDE/ILVR [8, 37]	104.81	0.676	16.27	0.318	100.54	0.665	16.62	0.290	226.51	0.784	10.69	0.345
MCG [10]	320.63	1.005	12.01	0.121	306.43	0.963	11.80	0.076	295.10	1.000	<u>11.62</u>	0.278
DPS [9]	98.02	0.517	17.35	0.364	<u>77.74</u>	<u>0.407</u>	<u>20.90</u>	<u>0.551</u>	111.73	0.569	10.66	<u>0.396</u>
DSG [42]	<u>87.51</u>	<u>0.375</u>	<u>20.75</u>	<u>0.524</u>	<u>107.25</u>	<u>0.453</u>	19.58	0.488	<u>110.85</u>	<u>0.546</u>	8.425	0.382
FGPS (Ours)	56.46	0.294	21.70	0.574	49.25	0.267	22.01	0.601	24.44	0.192	15.96	0.686

Table 1. Quantitative comparison of different methods on FFHQ and ImageNet validation datasets. Bolded entries indicate best performance and underlined entries indicate second best performance.

and generate images with fine-grained characteristics of the underlying image, it tends to also generate artifacts. In contrast, due to our frequency curriculum, our method is able to generate images with high perceptual quality. We give further qualitative comparisons to baselines in Appendix G.

Role of Dataset. For the ImageNet dataset, we observe that FGPS significantly outperforms all baselines, especially in perceptual quality. We conjecture that this is because on more complex datasets with varying frequency characteristics, the explicit frequency schedule for the measurement is essential to ensure the reverse process iterates do not deviate far from the intermediate noisy data manifolds learned by the diffusion models.

Results on Image Dehazing. Table 2 shows our main results for the nonlinear image dehazing task on the FFHQ dataset. A key strength of our method is the ability to handle nonlinear inverse problems, unlike methods such as [8, 24, 37]. In Figure 3, we observe that our method successfully reconstructs finer details such as background details and facial characteristics similar to what we observed for linear inverse problems. DSG produces slightly grainy

reconstructions, but performs much better on the dehazing task than the other tasks. Due to the powerful generative capacity of diffusion models, our method also performs better than problem-specific supervised methods such as AOD-Net [26], a phenomenon also observed in [35]. We provide more qualitative examples in Appendix G.

Method	FID↓	LPIPS↓	PSNR↑	SSIM↑
DoubleDIP [18]	50.51	0.500	13.80	0.483
AOD-Net [26]	33.35	0.325	17.42	0.532
DPS [9]	24.24	0.137	25.88	0.806
DSG [42]	<u>10.55</u>	<u>0.134</u>	<u>26.07</u>	<u>0.795</u>
FGPS (Ours)	9.37	0.066	26.88	0.840

Table 2. Comparison of different haze removal methods. Bolded entries indicate best performance and underlined entries indicate second best performance.

Visualizing Measurement Sequence. In Figure 4, we show the transformed measurements at three timesteps in the reverse diffusion process when running FGPS. In the early stages of the reverse process, the measurement retains

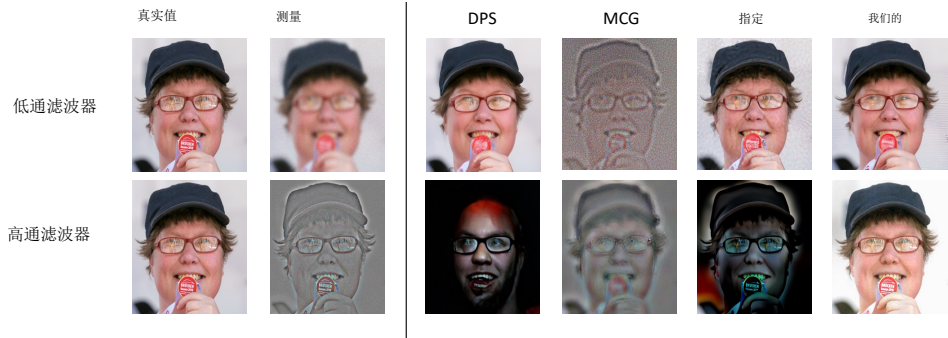


图2。即使对于线性逆问题，基于 SOTA 扩散的方法（DPS、MCG、DSG）在正向算子为高通滤波器卷积时，也可能难以求解逆问题。相比之下，我们的方法仍能实现高质量的重建。

方法	高斯模糊				动态模糊				高通滤波器			
	FID↓	LPIPS↓	PSNR↑	SSIM↑	FID↓	LPIPS↓	PSNR↑	SSIM↑	FID↓	LPIPS↓	PSNR↑	SSIM↑
FFHQ 数据集												
得分 SDE / ILVR [8 , 37]	58.79	0.266	23.34	0.648	69.10	0.387	18.59	0.457	85.35	151.77	0.356	0.581
MCG [10]	310.33	1.035	11.64	0.143	296.93	0.771	11.54	0.127	11.86	12.48	0.600	0.466
DPS [9]	21.69	0.116	26.01	0.743	27.15	0.153	24.17	0.688	121.57	0.694	6.17	0.222
DSG [42]	44.63	0.294	22.75	0.633	32.74	0.257	25.66	<u>0.687</u>	17.81	<u>0.246</u>	<u>0.124</u>	<u>12.12</u>
FGPS（我们的）	<u>21.89</u>	<u>0.139</u>	<u>25.44</u>	<u>0.727</u>	20.48	0.127	<u>25.48</u>	0.723	10.02	<u>0.695</u>	<u>0.739</u>	
ImageNet数据集												
评分- SDE / ILVR [8 , 37]	104.81	0.676	16.27	0.318	100.54	0.665	16.62	0.290	226.51	0.784	10.69	0.345
MCG [10]	320.63	1.005	12.01	0.121	306.43	0.963	11.80	0.076	295.10	1.000	<u>11.62</u>	0.278
DPS [9]	98.02	0.517	17.35	0.364	77.74	0.407	20.90	0.551	111.73	0.569	10.66	<u>0.396</u>
DSG [42]	87.51	0.375	20.75	0.524	<u>107.25</u>	<u>0.453</u>	<u>19.58</u>	<u>0.488</u>	110.85	0.546	8.425	0.382
FGPS（我们的）	56.46	0.294	21.70	0.574	49.25	0.267	22.01	0.601	24.44	0.192	15.96	0.686

表1.不同方法在 FFHQ 和ImageNet验证数据集上的定量比较。加粗条目表示最佳性能，下划线条目表示次优性能。

在生成具有底层图像细粒度特征的图像时，该方法往往也会产生伪影。相比之下，由于采用了频率课程，我们的方法能够生成具有高感知质量的图像。我们在附录G中提供了与基线方法的进一步定性比较。

数据集的作用。对于ImageNet数据集，我们观察到FGPS显著优于所有基线方法，特别是在感知质量方面。我们推测这是因为对于具有不同频率特性的更复杂数据集，显式频率调度对于确保反向过程迭代不会偏离扩散模型学习到的中间噪声数据流形至关重要。

图像去雾结果。表2展示了我们在FFHQ数据集上针对非线性图像去雾任务的主要结果。与[8、24、37]等方法不同，我们方法的关键优势在于能够处理非线性逆问题。如图3所示，我们的方法成功重建了背景细节和面部特征等更精细的细节，这与我们在线性逆问题中观察到的结果相似。DSG会产生轻微的颗粒感

重建任务表现平平，但在去雾任务上的表现远超其他任务。得益于扩散模型强大的生成能力，我们的方法甚至优于AODNet[26]等特定问题的监督方法，这一现象在[35]中也有体现。更多定性示例详见附录G。

方法	FID↓	LPIPS↓	PSNR↑	SSIM↑
DoubleDIP [18]	50.51	0.500	13.80	0.483
AOD -Net [26]	33.35	0.325	17.42	0.532
DPS [9] DSG [42]	24.24	0.137	25.88	0.806
FGPS（我们的）	10.55	0.134	26.07	0.795
	9.37	0.066	26.88	0.840

表2. 不同除雾方法的比较。加粗条目表示最佳性能，下划线条目表示次优性能。

测量序列可视化。图4展示了运行FGPS时逆向扩散过程三个时间点的转换测量结果。在逆向过程的早期阶段，测量结果保持



Figure 3. Qualitative results on FFHQ and ImageNet datasets. The dotted red boxes highlight areas of the image where our method results in higher-quality reconstructions than DPS and DSG. Zoomed in versions can be found in the Appendix.

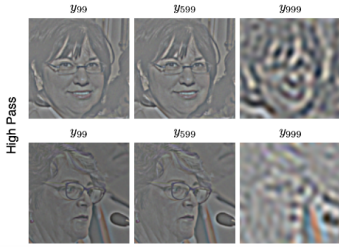


Figure 4. Visualization of Transformed Measurements y_t at different timesteps t , demonstrating our coarse-to-fine strategy.

only a coarse outline of the measurement. This coarse-to-fine strategy is essential to ensuring $\mathbf{x}_t$ remains on the data manifold learned by the diffusion model.

Effect of Frequency Curriculum. A key ingredient of our method is the frequency curriculum with a time-varying low pass filter $\mathbf{k}_t$. To understand the effect of this curriculum, we compare taking a time-dependent $\mathbf{k}_t$ vs. a fixed $\mathbf{k}$ that zeroes out frequencies above τ_1 for all t . In Figure 5, we see that for the high-pass filter operator, the generated reconstructions are still visually implausible images for a fixed $\mathbf{k}$, indicating the frequency curriculum is crucial to our method. Our method only requires $\mathbf{k}_1$ to be a low frequency cut-off, otherwise the reconstructions look like Figure 5. This corroborates our theoretical results, which show that the approximation gap early on in the reverse process greatly influences reconstruction quality. Finally, our

method does not qualitatively change with higher cutoffs for $\mathbf{k}_T$. We examine this in detail in Appendix G as well as conduct further ablation studies.



Figure 5. Qualitative Ablation Study: fixed low-pass filter vs. time-dependent low pass filter.

7. Conclusion

In this paper, we introduce Frequency Guided Posterior Sampling (FGPS), a method that progressively incorporates frequency information in a data-dependent manner. We provide a theoretical analysis characterizing the approximation error of FGPS and existing diffusion-based restoration methods, demonstrating how these errors can become severe when forward operators interact with high-frequency components of images. Our experiments on challenging image restoration tasks, such as motion deblurring and image dehazing, demonstrate that FGPS significantly outperforms existing methods. This work opens up new directions for analyzing diffusion-based inverse problem solvers by carefully considering the frequency characteristics of both the data and the degradation process.



图3. FFHQ 和ImageNet数据集的定性结果。红色虚线框标出了图像中我们的方法比DPS和DSG能生成更高质量重建的区域。放大版本见附录。

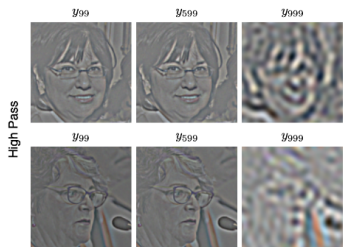


图4. 不同时间步 t 下变换测量值 y_t 的可视化，展示了我们的粗到细策略。

仅提供测量的粗略轮廓。这种从粗到细的策略对于确保 $\mathbf{x}_t$ 始终位于扩散模型学习到的数据流形上至关重要。

频率课程的影响。 我们方法的关键要素是带有时变低通滤波器 $\mathbf{k}_t$ 的频率课程。为理解该课程的影响，我们比较了采用时变 $\mathbf{k}_t$ 与固定 $\mathbf{k}$ （对所有 t 使频率高于 ω_1 的频率归零）的情况。在图5中，我们发现对于高通滤波算子，当 $\mathbf{k}$ 固定时，生成的重建图像在视觉上仍显得不真实，这表明频率课程对我们的方法至关重要。我们的方法仅需 $\mathbf{k}_1$ 为低频截止值，否则重建图像如图5所示。这验证了我们的理论结果，即逆过程早期的近似间隙对重建质量影响显著。最后，我们的

该方法不会因 $\mathbf{k}_T$ 的截断值提高而发生定性变化。我们将在附录G中对此进行详细分析，并开展进一步的消融研究。



图5. 定性消融研究：固定低通滤波器与时间依赖性低通滤波器对比。

7. 结论

本文提出频率引导后验采样（FGPS）方法，该方法能以数据依赖的方式逐步整合频率信息。我们通过理论分析揭示了FGPS及现有扩散式图像修复方法的近似误差特征，证明当前算子与图像高频成分相互作用时，这些误差可能急剧放大。在运动模糊消除、图像去雾等高难度修复任务的实验中，FGPS展现出显著优于现有方法的性能优势。本研究通过系统考量数据与退化过程的频率特性，为扩散式逆问题求解器的分析开辟了新方向。